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Transparent Menu

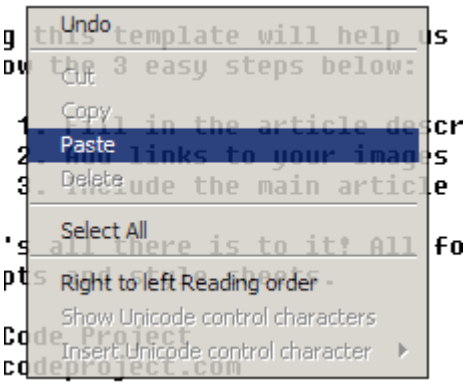
Derick Cyril Thomas

8 Jun 2004

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An article on creating transparent menus for your applications

Download demo project - 142.2 KB



Introduction

Mac OS has nice transparent menus, which uses hardware acceleration. It is almost impossible to achieve that kind of effect in MS Windows, but one can surely create menus, which are looking like that of Mac. The topic of this article is not how to create Mac like menus, but discuss some of the options available for a Windows developer to create nice owner drawn menus. Here in this article, I present four different approaches, to implement a transparent menu.

- Using `WM_INITMENUPOPUP` message
- Using Owner Draw method
- Superclassing menu class

- Using Hooks

All versions of Windows from Win2K allow you ... provides no API which returns window HANDLE ... to get a handle to menu and set transparency.

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Setting Transparency for a

Windows 2000 and above provides a function ...

C++

```
BOOL SetLayeredWindowAttributes(HWND hwnd, ...
```

This function can be used in the following manner to set transparency for a window given its HWND.

C++

```
//Setting transparency for a normal window
DWORD dwStyle = GetWindowLong(hWnd, GWL_EXSTYLE);
//WS_EX_LAYERED Flag must be specified with extended styles
SetWindowLong(hWnd, GWL_EXSTYLE, dwStyle|WS_EX_LAYERED);
SetLayeredWindowAttributes(hWnd, 0, 150, LWA_ALPHA);
  //150 represent transparency level
```

The transparency level can vary from 0 to 255. Zero means completely transparent (invisible) and 255 corresponds to fully opaque.

Making of a Transparent Menu

There occurs some complication in the case of menus. In addition to transparency feature MS has also provided Menu animations, which will interfere with transparency settings, if you apply the above code to menu window. So, as a workaround, we will temporarily disable menu animations with the following code:

C++ Shrink ▲

```
//Disable menu animation
OSVERSIONINFO ovi;
ovi.dwOSVersionInfoSize = sizeof(OSVERSIONINFO);
BOOL bRet = GetVersionEx(&ovi);
// Toggle menu fading/animation mode (Windows2K or later)
BOOL bMenuFading = FALSE;
BOOL bMenuAnim = FALSE;
if( ovi.dwMajorVersion >= 5 ){
  SystemParametersInfo(SPI_GETMENUFADE, 0, &bMenuFading, 0);
  SystemParametersInfo(SPI_GETMENUANIMATION, 0, &bMenuAnim, 0);
  SystemParametersInfo(SPI_SETMENUFADE, 0,
```

```
    (LPVOID) FALSE, SPIF_SENDWININICHANGE);
  SystemParametersInfo(SPI_SETMENUANIMATION, 0,
    (LPVOID) FALSE, SPIF_SENDWININICHANGE);
}

//Transparency code goes here...

if( ovi.dwMajorVersion >= 5 ){
  SystemParametersInfo(SPI_SETMENUFADE, 0,
    (LPVOID) bMenuFading, SPIF_SENDWININIC
  SystemParametersInfo(SPI_SETMENUANIMATIC
    (LPVOID) bMenuAnim, SPIF_SENDWININICHA
}
```

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I will be omitting the above code fragment in t
should be there for proper working.

1. Handling WM_INITMENUPOPUP

In this method, we will be handling `WM_INITMENUPOPUP` message. From MSDN, "The `WM_INITMENUPOPUP` message is sent when a drop-down menu or submenu is about to become active. This allows an application to modify the menu before it is displayed, without changing the entire menu". What is done is that, when `WM_INITMENUPOPUP` is generated, the menu will already be created. So we will be using `FindWindow()` API to get `HWND` of the menu. Then we will be modifying menu, to suit our purpose. Class name for menu is `#32768`. Here is the code:

C++ Shrink ▲

```
//Main window message handler procedure
LRESULT CALLBACK WndProc(HWND hWnd, UINT message,
  WPARAM wParam, LPARAM lParam)
{
  switch (message)
  {
    case WM_INITMENUPOPUP:
    {
      HWND hMenuWnd = FindWindow(_T("#32768"), NULL);
      if (IsWindow(hMenuWnd)) {
        DWORD dwStyle;
         dwStyle = GetWindowLong(hMenuWnd, GWL_EXSTYLE);

        dwStyle = SetWindowLong(hMenuWnd, GWL_EXSTYLE,
          dwStyle|WS_EX_LAYERED);
        SetLayeredWindowAttributes(hMenuWnd, 0, 150, LWA_ALPHA);
      }
    }
    break;

    case WM_CONTEXTMENU:
    {
      HMENU hMenu = LoadMenu(hInst, MAKEINTRESOURCE(IDC_TMENU1));
      if (IsMenu(hMenu)) {
        DWORD dwPos = ::GetMessagePos();
```

```
        POINT pt = {LOWORD(dwPos), HIWORD(dwPos)};
        HMENU hSubMenu = GetSubMenu(hMenu, 0);
        TrackPopupMenuEx(hSubMenu, TPM_LEF
          TPM_RIGHTBUTTON | TPM_NOANIMATIC
          pt.x, pt.y, hWnd, NULL);
      }
    }
    break;
//Other handlers go here...
    default:
      return DefWindowProc(hWnd, message,
  }
  return 0;
}
```

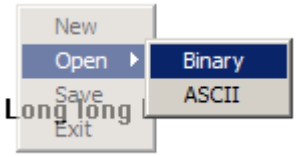
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Note that this technique only works with top-le individual controls. During my testing, I got `WM_INITMENUPOPUP` for all menus, but `SetWindowLong()` failed for all menus except Context Menu, even popup menus inside a context menu also failed. A screen shot is given below:



So its application is limited to context menu like purpose. Another point worth noting is that menu animations can be turned off without using `SystemParametersInfo()`. This is because `TrackPopupMenu()` / `TrackPopupMenuEx()` can take a parameter `TPM_NOANIMATION` which will prevent menu animations. You can use this method effectively in applications that require only simple context menus and button menus. Attached project `tMenu1` illustrates this method.

2. Owner-Drawn Menu

If you have an owner-drawn menu, then this method is for you. When we receive `WM_DRAWITEM` message, we will also be receiving HDC to menu. So we can obtain handle of the menu window using `WindowFromDC()` API. This call will not always return a valid handle, because of naughty menu animations. Windows provides a temporary DC for menu animations. But once we get a valid DC, we can perform all sorts of operations we require on that menu window. So you need to set `MF_OWNERDRAW` style set to each menu item.

C++ Shrink ▲

```
LRESULT CALLBACK WndProc(HWND hWnd,
   UINT message, WPARAM wParam, LPARAM lParam)
{
  switch (message)
  {
    case WM_MEASUREITEM:
    {
      LPMEASUREITEMSTRUCT lpms = (LPMEASUREITEMSTRUCT)lParam;
```

```
      lpms->itemWidth = 80;
      lpms->itemHeight=40;
    }
    return 0;
    case WM_DRAWITEM:
    {
      LPDRAWITEMSTRUCT lpdis = (LPDRAWITEM
      HWND hMenuWnd = WindowFromDC(lpdis->
      if (IsWindow(hMenuWnd)) {
        DWORD dwStyle =0;
        dwStyle = GetWindowLong(hMenuWnd,
        SetWindowLong(hMenuWnd, GWL_EXSTYL
        SetLayeredWindowAttributes(hMenuWr
      }
      //TODO:Rest of the drawing stuff...
    }
    return 0;
    case WM_CONTEXTMENU:
    {
      HMENU hMenu = CreateMenu();
      HMENU hMenuPopup = CreatePopupMenu();
      AppendMenu(hMenuPopup, MF_STRING|MF_OWNERDRAW,
          ID_FILE_NEW, (LPCTSTR)"&Pub");
      AppendMenu(hMenuPopup, MF_STRING|MF_OWNERDRAW,
          ID_FILE_OPEN, (LPCTSTR)"&Club");
      AppendMenu(hMenu, MF_POPUP|MF_OWNERDRAW,
          (UINT)hMenuPopup, (LPCTSTR)"&labamba0");
      DWORD dwPos = ::GetMessagePos();
      POINT pt = {LOWORD(dwPos), HIWORD(dwPos)};
      TrackPopupMenuEx(hMenuPopup, TPM_LEFTALIGN | TPM_RIGHTBUTTON
          |TPM_NOANIMATION, pt.x, pt.y, hWnd, NULL);
      DestroyMenu(hMenu);
    }

    return 0;
//Rest of message handler
  }//switch
}
```

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Here also some glitches occur, which are of course caused by menu animations. You have to hover mouse pointer over menu to make it transparent. To overcome this bad effect, you should specify `TPM_NOANIMATION` with `TrackPopupMenuEx()` flags. But it still has got a bit of problem with popup menus. Here's a screenshot:



This method can be very easily integrated to owner-drawn menus. It can give nice transparent effect without too much tedious work. Attached project `tMenu0` illustrates this method.

3. Superclassing Menu

This is one of those rare methods. I haven't see
this method, we obtain information about mer
replace its window procedure with our one and
Hence, all the menus created afterwards will be
menu is #32768. The following code illustrates

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C++

```
//Superclassing menu

LRESULT CALLBACK MenuWndProc(HWND, UINT, W

int APIENTRY _tWinMain(HINSTANCE hInstance,
    HINSTANCE hPrevInstance,
    LPTSTR lpCmdLine,int nCmdShow)
{
  WNDCLASS wc;

  GetClassInfo(NULL, _T("#32768"), &wc);
  wc.hInstance = hInstance;
  wpOldMenuProc = wc.lpfnWndProc;
  wc.lpfnWndProc = (WNDPROC)MenuWndProc;
  SetLastError(0);
  RegisterClass(&wc);

  //Rest of the stuff goes here...
  return 0;
}
```

Now MenuWndProc can be written as follows:

C++

```
//Message Handler for Menu
LRESULT CALLBACK MenuWndProc(HWND hWnd, UINT uMsg,
  WPARAM wParam, LPARAM lParam)
{
switch(uMsg) {
case WM_CREATE:
  {
  LRESULT lRet = CallWindowProc(wpOldMenuProc,
    hWnd, uMsg, wParam, lParam);
  DWORD dwStyle = GetWindowLong(hWnd, GWL_EXSTYLE);
  dwStyle &= ~WS_EX_WINDOWEDGE;
  dwStyle &= ~WS_EX_DLGMODALFRAME;
  SetWindowLong(hWnd, GWL_EXSTYLE, dwStyle|WS_EX_LAYERED);
  SetLayeredWindowAttributes(hWnd, 0, 180, LWA_ALPHA);
  return lRet;
  }
default:
  return CallWindowProc(wpOldMenuProc, hWnd, uMsg, wParam, lParam);
}//switch
```

```
return 0;
}
```

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This method works great for all kind of menus. Popup menus. But as usual, the "*Fade Effect*" ar causes interference with menus. Attached sam

4. Using Windows Hooks

Hooks represent one of those powerful abilitie be monitored using Hooks. As MSDN says, the can mess things up. By "hooking", you tell Win procedure, that will be called every time an eve them: ***local*** and ***remote*** hooks. We are interested in local hooks, as they will trap events occurring in local thread.

When you create a hook, Windows creates a data structure in memory, containing information about the hook, and adds it to a linked list of existing hooks. New hook is added in front of old hooks. When an event occurs, if you install a local hook, the filter function in your process is called so it's rather straightforward. One can install Hooks with `SetWindowsHookEx()` function. You can uninstall a hook by calling `UnhookWindowsHookEx()` which accepts only one parameter, the handle of the hook you want to uninstall. When an event occurs, Windows calls `HookProcedure`.

We will be using `WH_CALLWNDPROC` which is initiated when a `SendMessage()` is called. Then we will check if the window is a menu window by checking its class name. If it is so, we will proceed with our transparency operation. The following code illustrates this:

C++ Shrink ▲

```
//Main window Procedure
LRESULT CALLBACK WndProc(HWND hWnd, UINT message,
  WPARAM wParam, LPARAM lParam)
{
  int wmId, wmEvent;
  static HWND    hWndEdit;
  static HHOOK  hHook;
  switch (message)
  {
  case WM_CREATE:
    hHook = SetWindowsHookEx(WH_CALLWNDPROC, HookCallWndProc,
       0,GetWindowThreadProcessId(hWnd,0));
    hWndEdit = CreateWindowEx(WS_EX_CLIENTEDGE, _T("EDIT"),NULL,
       WS_CHILD | WS_VISIBLE | ES_MULTILINE | WS_HSCROLL | WS_VSCROLL,
       0,0,0,0,hWnd,(HMENU)NULL,hInst,NULL;
    SendMessage(hWndEdit, WM_SETFONT, (
      WPARAM)GetStockObject(SYSTEM_FIXED_FONT), MAKELPARAM(TRUE, 0));
    break;
  case WM_DESTROY:
    if(hHook!=0)
      UnhookWindowsHookEx(hHook);
```

```
    PostQuitMessage(0);
    break;
  default:
    return DefWindowProc(hWnd, message, wP
  }
  return 0;
}

LRESULT CALLBACK HookCallWndProc(int nCode
  WPARAM wParam, LPARAM lParam)
{
  CWPSTRUCT cwps;
  LONG lRet;
  if( nCode == HC_ACTION ){
    CopyMemory(&cwps, (LPVOID)lParam, size
  switch(cwps.message){

  case WM_CREATE:
    {

      CHAR szClass[128];
    GetClassName(cwps.hwnd, szClass, 127);
    if(_tcscmp(szClass, _T("#32768"))==0){
      HWND hMenuWnd = cwps.hwnd;
      DWORD dwStyle =0;
      dwStyle = GetWindowLong(hMenuWnd, GWL_EXSTYLE);
      SetWindowLong(hMenuWnd, GWL_EXSTYLE, dwStyle|WS_EX_LAYERED);

      SetLayeredWindowAttributes(hMenuWnd, 0, 220, LWA_ALPHA);
    }
    }
    break;
   }//switch
  }//if
  return CallNextHookEx((HHOOK)WH_CALLWNDPROC, nCode, wParam, lParam);
}
```

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The hooks are chained in a linked list with the most recently installed hook at the head of the list. When an event occurs, Windows will call only the first hook in the chain. It's your hook procedure's responsibility to call the next hook in the chain. You can choose not to call the next hook but you'd better know what you're doing. Most of the time, it's a good practice to call the next procedure so other hooks can have a shot at the event. You can call the next hook by calling `CallNextHookEx()`

As you have seen, it is a slightly difficult technique. But it is very useful in the sense that very window can be caught in the `HookWindow` Procedure. For more information on Windows Hooks, you can visit Iczelion's Win32 Assembly Homepage, although it is in assembly language it is pretty useful. This method is illustrated in the attached `tMenu3` sample project. The fist screenshot applies to the last two techniques.

Which One to Choose...

The answer to this question entirely depends upon your application. If you have simple applications, like one that is in a dialog box, then you can ch / of the last two methods to get support for eve u should be very careful and must unhook when level, you can have your applications' nice look much transparency can prevent users from see 0 and 245.

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Concluding Words...

When this article was first posted, it only inclu more methods. I have included all those sugge suppose that this article has something worthwhile. Thank you all for your valuable suggestions and they are welcome.

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get rid of menu animations
seddi 12-Aug-14 13:33

Neat
KevinSW 27-Jul-14 17:43

broken link
JJMatthews 22-Aug-12 22:55

My vote of 5
JJMatthews 22-Aug-12 22:37

Works as suggested.
parandor 9-Jun-11 6:36

Superclassing System MENU #32768 and other system provided classes
manurocks 11-Jun-10 7:55

WM_ENTERIDLE [modified]
cantstandya 1-Jun-10 11:37

placing custom button in another application's title bar.
Member 4140119 8-Dec-08 21:12

transparecy over video
Saiz 21-Jun-07 2:22

Problem with Owner-Drawn method
AnandChavali 27-Oct-06 0:50

Re: Problem with Owner-Drawn method
AnandChavali 13-Nov-06 19:50

#32768
Sarath C 12-Oct-06 19:00

Re: #32768
AnandChavali 13-Nov-06 19:45

when i use right mouse button click caption or edit,show a button on taskbar,why?
user8848 30-Aug-06 17:39

why SetLayeredWindowAttributes can't used in 8 bits(256)
Eureka Jim 11-Aug-04 12:01

Impressive Work but bug
ChetnaBABY 29-Apr-04 18:56

(255 / 100) * (100 - 0) = 255
.:floyd:. 6-Feb-04 15:32

Adding items to this popup menu
Parag Paithankar 5-Nov-03 22:08

Very convincing...
dandy72 20-Aug-02 2:24

Re: Very convincing...
.:floyd:. 4-Feb-04 9:04

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This could be done without hooks, i guess
10-Jul-02 10:27

Doesn't work on Windows Me
rushfn 18-Jun-02 9:17

Re: Doesn't work on Windows Me
Derick Cyril Thomas 21-Mar-03 2:36

Re: DB on Win 2k
11-May-02 17:31

It doesn't work on Win98.
1-May-02 12:48

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